

Muhammad Shoaib

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EDUCATION

University of Genoa

Genoa, Italy

PhD Researcher Supervisors: Prof. Eva Riccomagno

December 2023 – Present

- **Project:** Coherent Inference for Graphical Models with Applications to Risk Assessment of Rare Events

University of Valencia

Valencia, Spain

Visiting Researcher Supervisors: Prof. Gherardo Varando

February 2025 – June 2025

- **Projects:** Estimating Staged Event Tree Models via Hierarchical Clustering on the Simplex Events
- Robust Estimation of Country-Level Crop Yield at a Global Scale

Institute of Space Technology

Islamabad, Pakistan

Master of Science in Remote Sensing and GISc

Sep. 2014 – Aug 2018

- **Thesis:** Building extraction from high resolution satellite imagery using deep learning.

Quaid-I-Azam University

Islamabad, Pakistan

Master of Science in Statistics

Feb. 2010 – Feb 2012

- **Project:** Survey on quality of hospital information system in Islamabad and Abbottabad.

Hazara University

Mansehra, Pakistan

Bachelor of Science in Mathematics and Statistics

Sep. 2006 – Aug 2008

EXPERIENCE

Subject Specialist Statistics

May 2015 – Jan 2022

KPESE

Abbottabad, Pakistan

- Teaching Statistics to intermediate students

Data Science Enthusiast

Sep. 2014 – Dec 2022

Fiverr, Freelancer

- Worked on multiple project relating to data science, machine learning, remote sensing and GISc and Statistics etc. Some of the projects are:
 - * Depressive feelings of heart and cancer patients
 - * Using physical functioning, role-physical and bodily pain to predict the overall health score (QWBSCORE) using the National Health Measurement Study (NHMS) data.
 - * Comparision of SVM and Neural Net on Caravan data using IBM SPSS Modeler.

Business Intelligence Analyst

Dec 2013 – March 2014

360 Technologies

Islamabad, Pakistan

- Build churn prediction models for telecommunication companies(Zong and Telenor) using machine learning algorithms

Data Scientist

Feb 2013 – Nov 2013

Protege Global

Islamabad, Pakistan

- Working on Medical Lien Management, California, USA projects, performing data analysis on medical insurance settlements
- Making exploratory and predictive models for insurance claims and settlements
- Making regular analysis reports using graphical aides

CLASS PROJECTS

Earth Science <i>MATLAB, ENVI, R</i> • Time Series Analysis on Oceanographic Tide Data	Dec 2015 – Feb 2016
GML <i>MATLAB, Envi, Python Erdas Imagine</i> • Glaciers Monitoring using Landsat images	Feb 2015 – June 2015
MLC <i>MATLAB, Erdas Imagine</i> • Maximum Likelihood Classification on High Resolution Satellite Imagery	Oct 2014 – Dec 2014
QHIS <i>SPSS, Stata</i> • Survey on Quality of Hospital information system • Used frequency tables, graphical representation and Regression Analysis etc	Aug 2010 – Dec 2010

TECHNICAL SKILLS

Languages: MATLAB, Python, Basic HTML/CSS, R, Latex
RS and GIS Softwares: Erdas Imagine, QGIS, ArcGIS, Envi
Statistical Softwares: IBM SPSS, Stata, Minitab, SAS, Statistica

ONLINE COURSES

Python Data Structures: Coursera
Machine Learning: Coursera
Programming for Everybody (Getting Started with Python): Coursera
Data Management and Visualization: Coursera
Introduction to Big Data : Coursera
The Data Scientists Toolbox: Coursera
The Analytics Edge: edX
Data Science and Machine Learning Essentials: edX
Statistical Thinking for Data Science and Analytics: edX

RESEARCH INTERESTS

Probabilistic Graphical Modles, Information Geometry, Deep Learning, Data Science, Photogrammetry, Computer Vision, Statistical Learning, Remote Sensing and GISc

JOURNAL PUBLICATIONS

ASME 2020: Zubair, Muhammad, Sajid Ghuffar, **Muhammad Shoaib**, Ahmed Bilal Awan, and Abdul Rauf Bhatti. "Assessment of Photovoltaic Capabilities in Urban Environments: A Case Study of Islamabad, Pakistan." *Journal of Solar Energy Engineering* 142, no. 6 (2020)

MDPI Entropy 2023: Giovanni Pistone, **Muhammad Shoaib**, "Kaniadakis's Information Geometry of Compositional Data <https://doi.org/10.3390/e25071107>

MDPI Axioms 2024: Giovanni Pistone, **Muhammad Shoaib**, "A Unified Approach to Aitchison's, Dually Affine, and Transport Geometries of the Probability Simplex <https://doi.org/10.3390/axioms13120823>

CONFERENCE PUBLICATIONS

ICASE 2015: Muhammad Shoaib, Muhammad Qasim, Zaka Ur Rehman "Comparison of Maximum Likelihood Classification Before and After Applying Weierstrass Transform" in Fourth International conference on Aerospace Science and Engineering, page 289, 2015

IEEEExplore 2013: Aftab, S., Abbas, W., Bilal, M. M., Hussain, T., **Shoaib, M.**, & Mehmood, S. H. (2013, September). "Data mining in insurance claims (DMICS) twoway mining for extreme values." In Eighth International Conference on Digital Information Management (ICDIM 2013) (pp. 1-6). IEEE

POSTER

FOSS4G 2018: Muhammad Shoaib, S Ghuffar. "Building Extraction In High Resolution Satellite Imagery Using Deep Learning" in FOSS4G Asia 2018

4th Italian Meeting on Probability and Mathematical Statistics 2024: Muhammad Shoaib, Eva Riccomagno . "Exploring and Applying Divergence Measures and Distances in the Analysis of Compositional Data and Their Real-World Applications"

CURRENT PROJECTS

"Estimating Staged Event Tree Models via Hierarchical Clustering on the Simplex", currently in its final stages. This work is a collaboration with Prof. Eva Riccomagno (UNIGE), Prof. Gherardo Varando and Prof. Manuele Leonelli (IE University).

Robust estimation of country-level crop yield at a global scale. This research originated from a week-long hackathon at the ISP Lab and is being conducted in collaboration with researchers from the ISP group (UVEG) and OrionLab (NTUA-NOA, Athens, Greece).

SPOKEN LANGUAGES

Urdu: National Language

Hindko: Mother tongue

English: B2

SPORTS AND HOBBIES

Hockey: Part of Departmental Winner of Hockey Team at Quaid-I-Azam University, Islamabad(Played as right full back)

Cricket Played Cricket From the Statistics Department at Quaid-I-Azam University, Islamabad

Calisthenics Athlete

REFREES

Prof. Eva Riccomagno: eva.riccomagno@unige.it
Univerisyt of Genova, Italy

Prof. Gherardo Varando: gherardo.varando@uv.es
University of Valencia, Spain